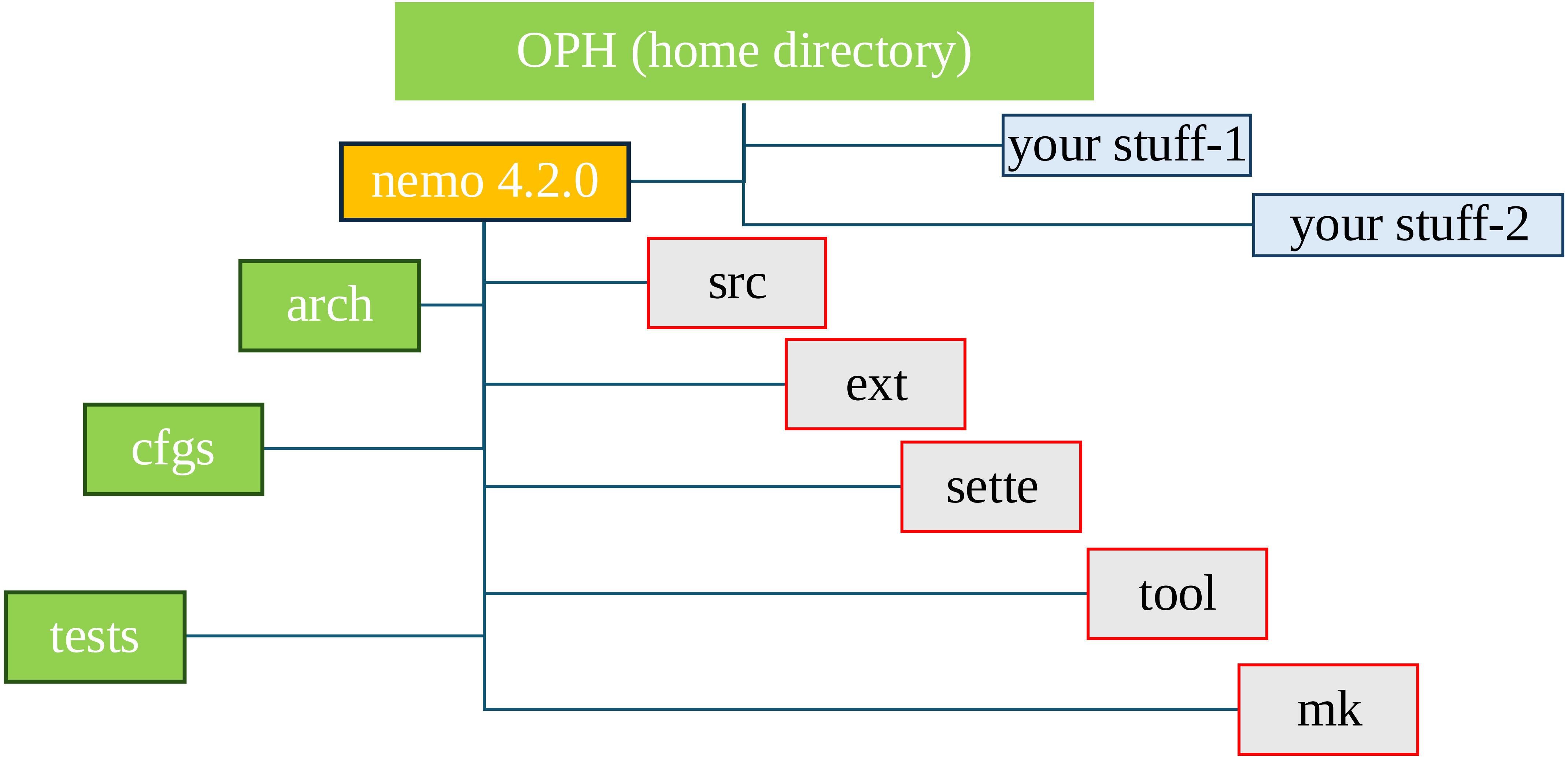
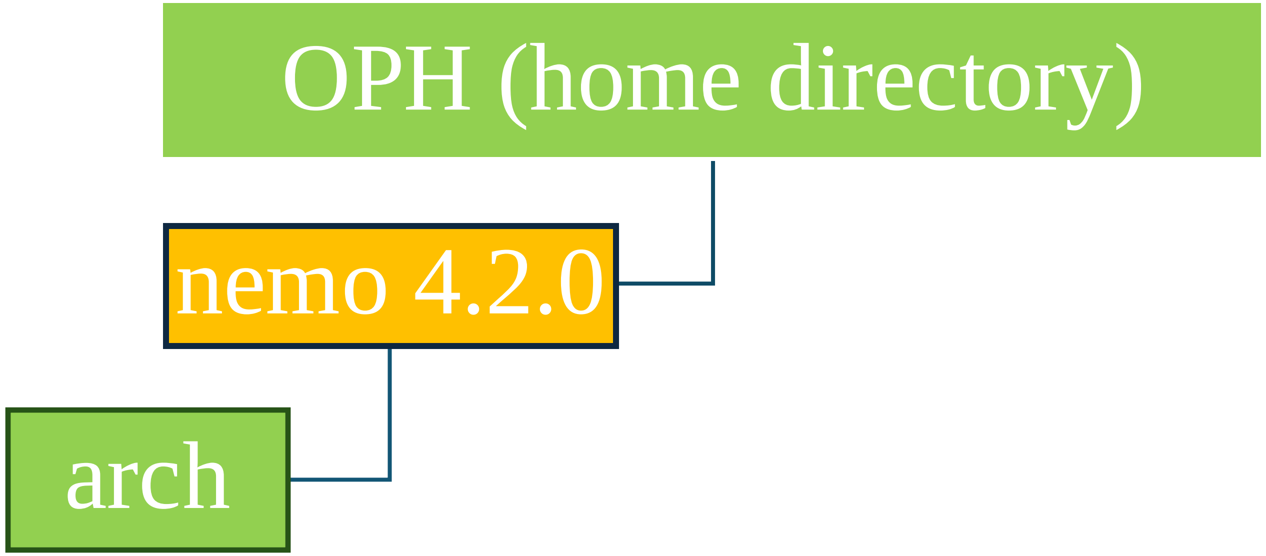
NEMO architecture

NEMO architecture



No need to work on these directories

NEMO architecture

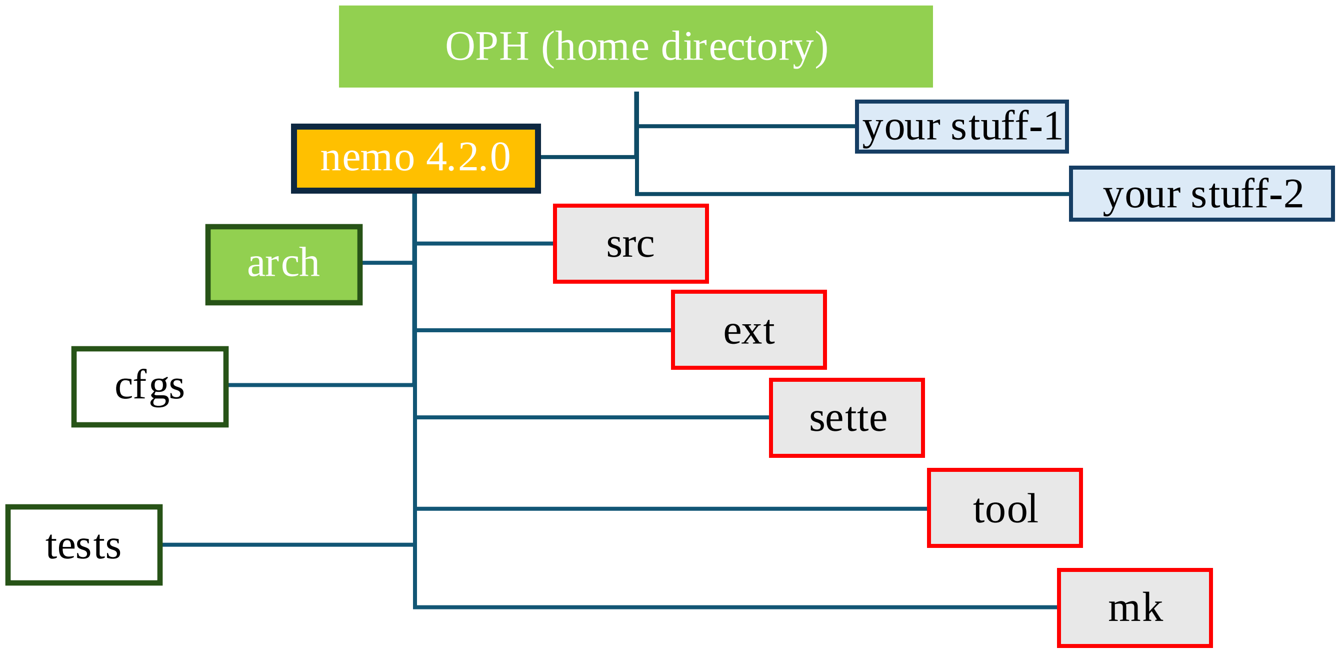


This directory contains the compiling options and a description of the HPC architecture.
Only 1 action is needed is this directory.
Once you are in this directory, copy the OPH specifc file here

> cp /home/software/terra/NEMO_env/arch_nemo/arch-OPH.fcm .

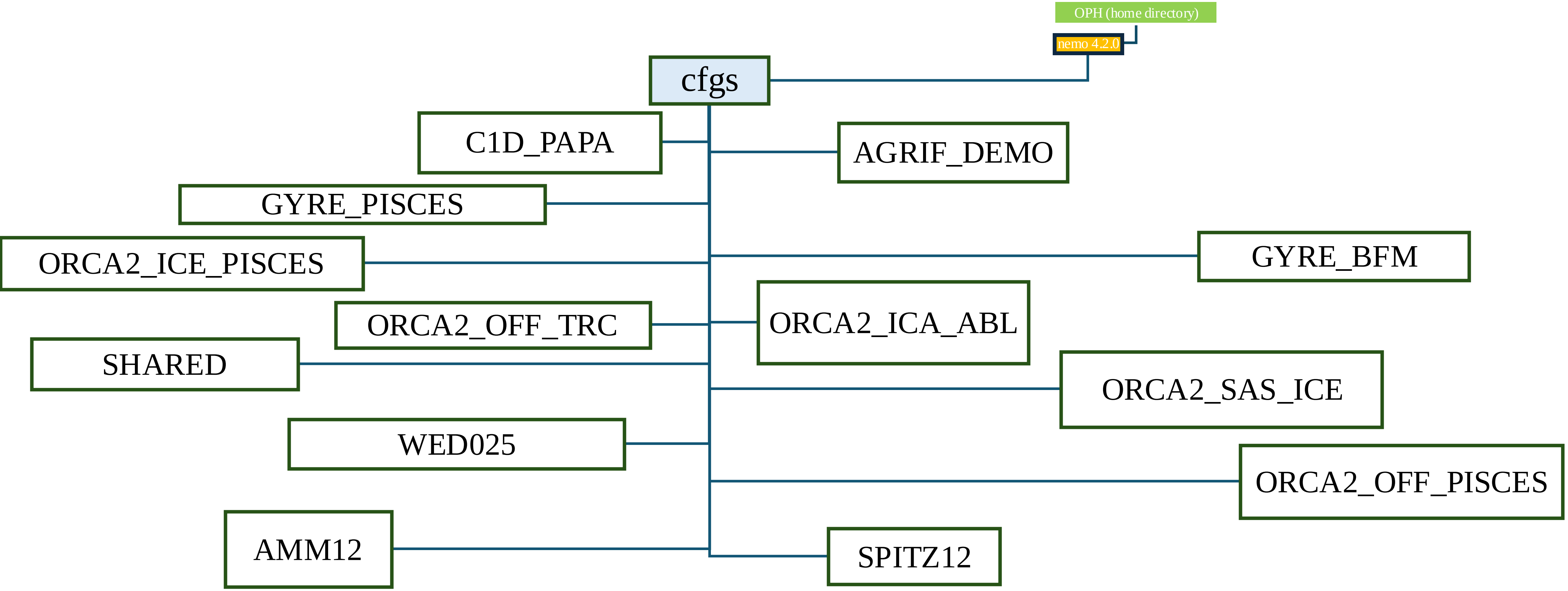
NEMO architecture

The approach for each experiment you want to perform is to start from a pre-existing experiment.
There are 2 categories of pre-existing experiments.
Configuration (CFGS)
Test cases (TESTS)



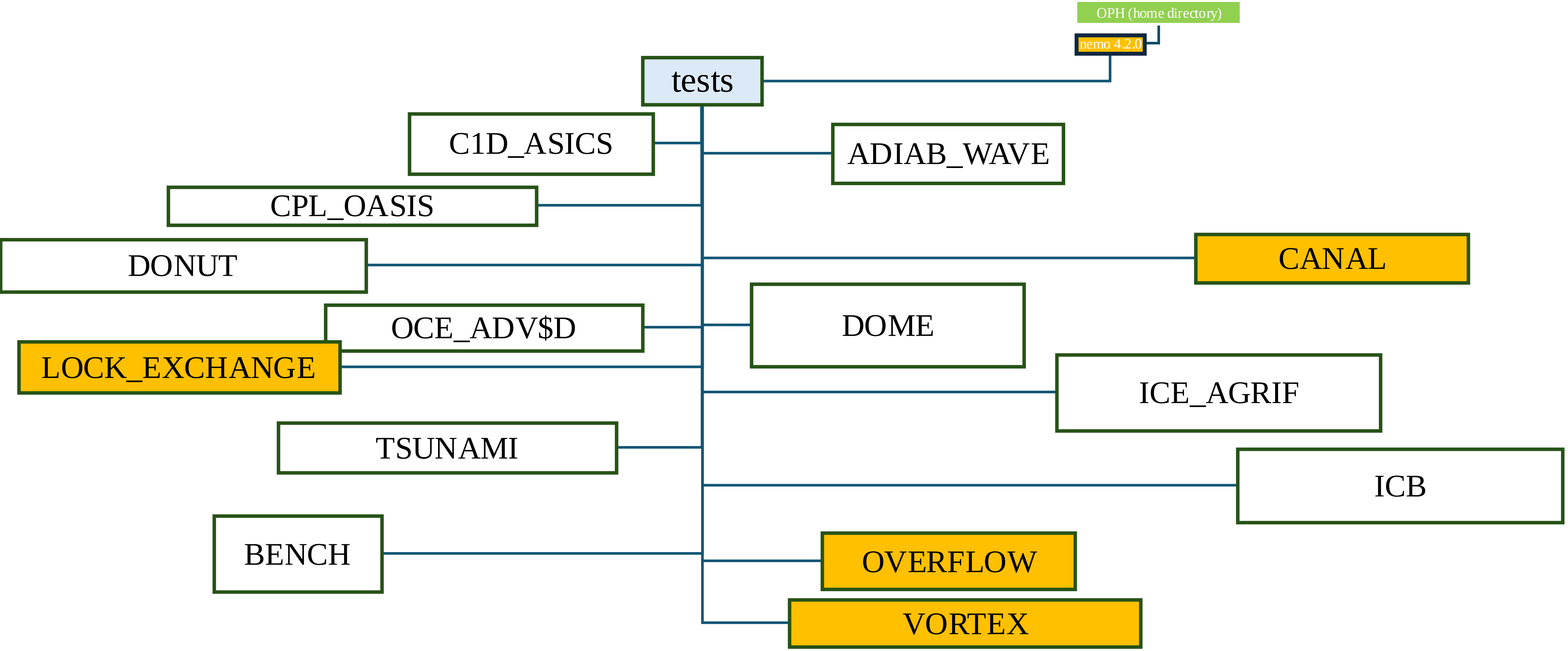
NEMO architecture

Configuration (CFGS): are realistic implementations of NEMO



NEMO architecture

Tests (tests): are idealized implementations of NEMO



NEMO architecture

To create your own experiment

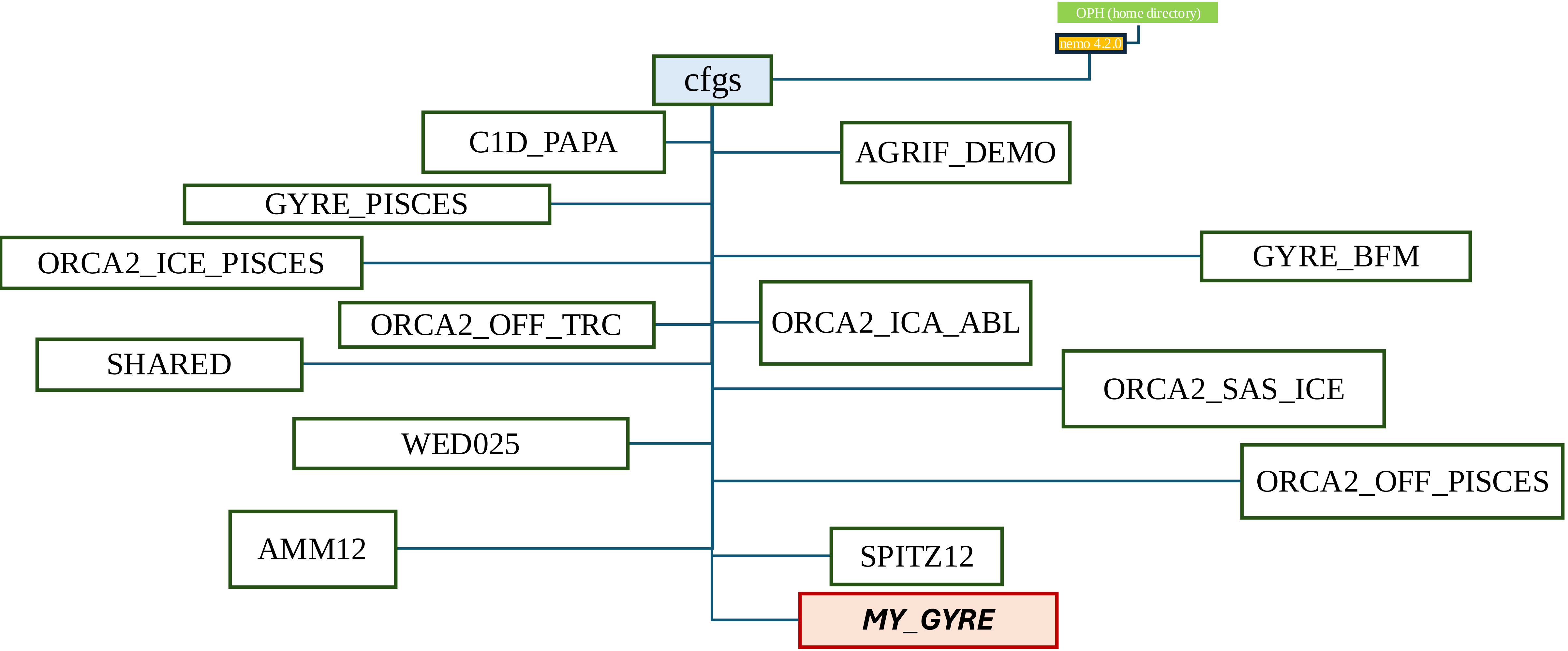
- ./makenemo -m 'OPH' -r GYRE_PISCES -n 'MY_GYRE’
- ./makenemo -m 'OPH' -a VORTEX -n 'MY_VORTEX’

-r: you clone a pre-existing nemo configuration setup

-a: you clone a pre-existing nemo test case setup

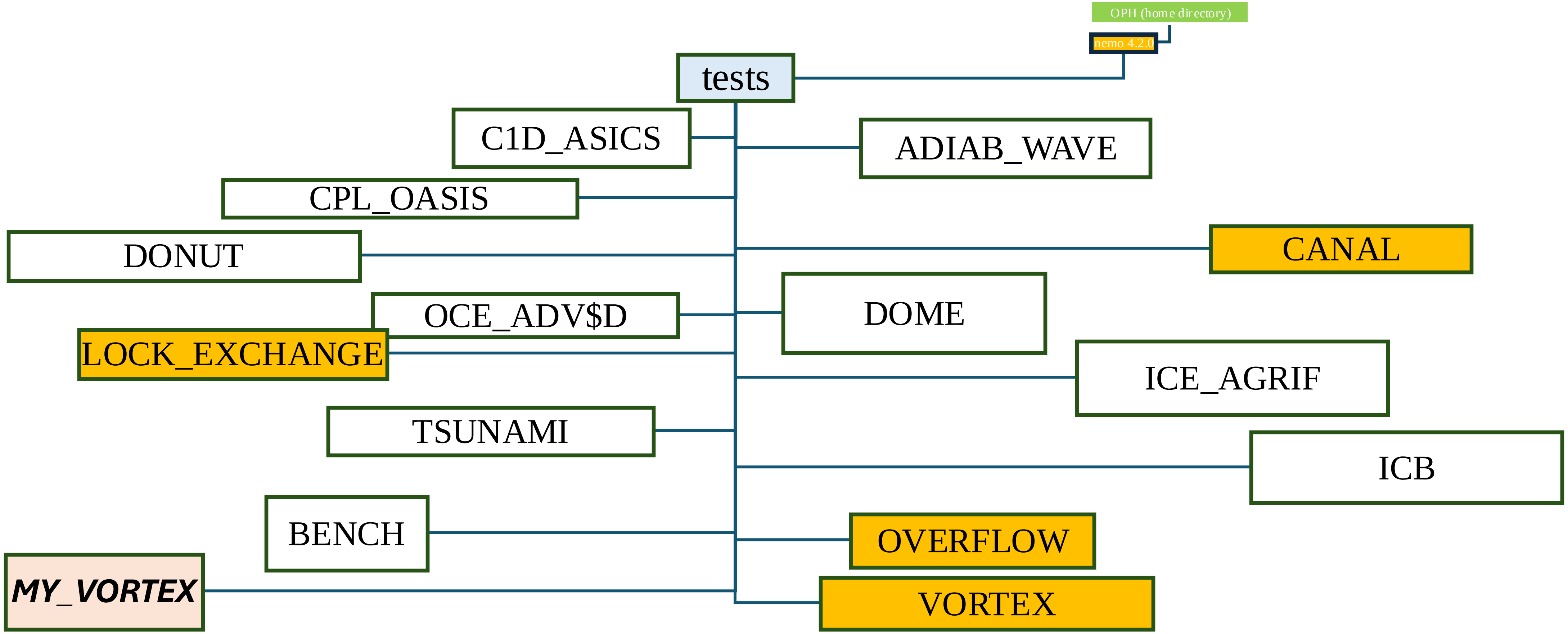
NEMO architecture

If you created you experiment from a pre-existing configuration



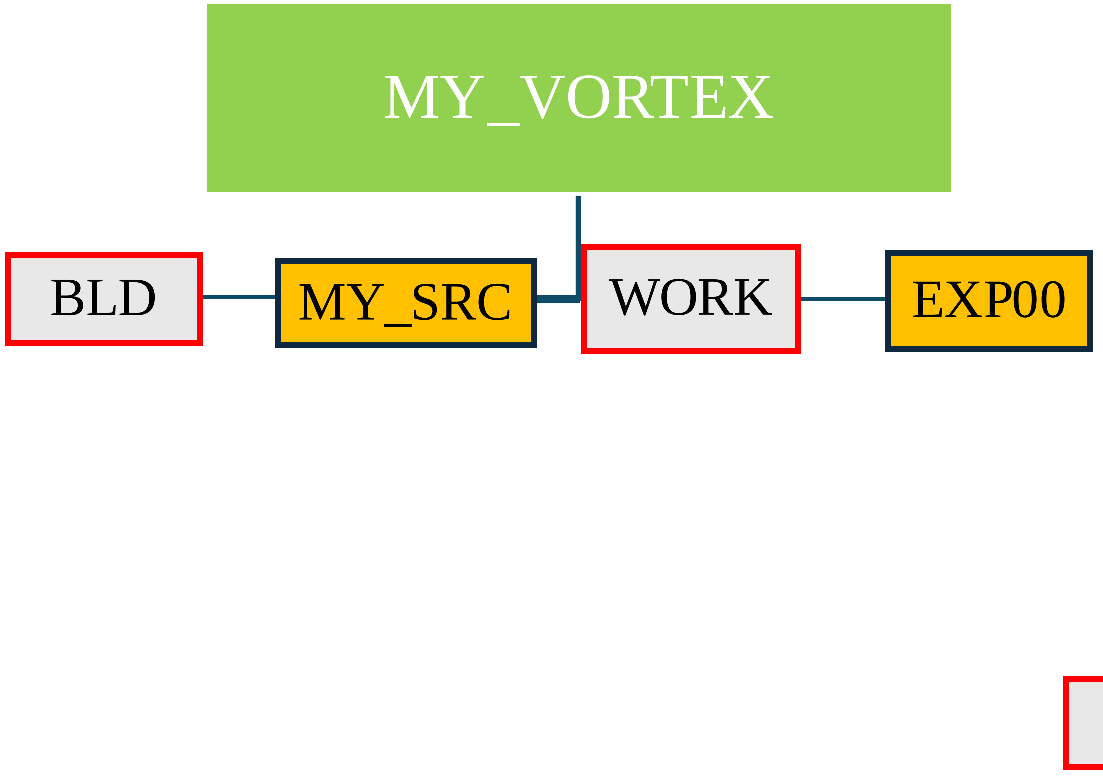
NEMO architecture

If you created you experiment from a pre-existing test



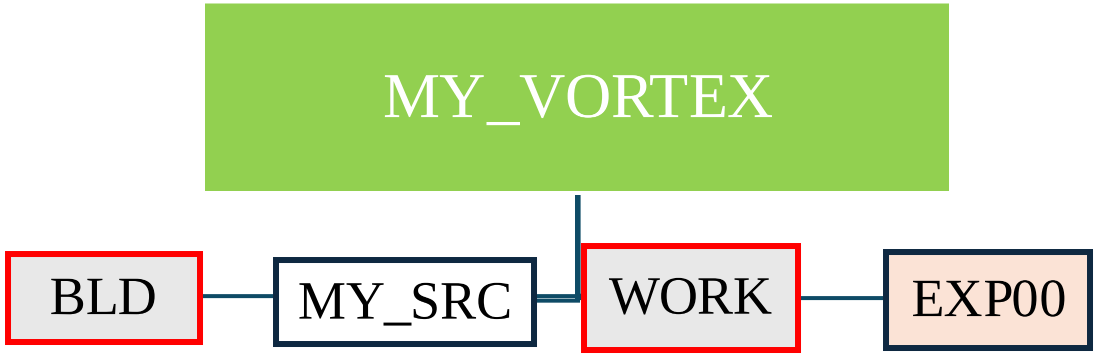
NEMO architecture

In both cases the directories structures within your experiment is the same



No need to work on these directories

NEMO architecture

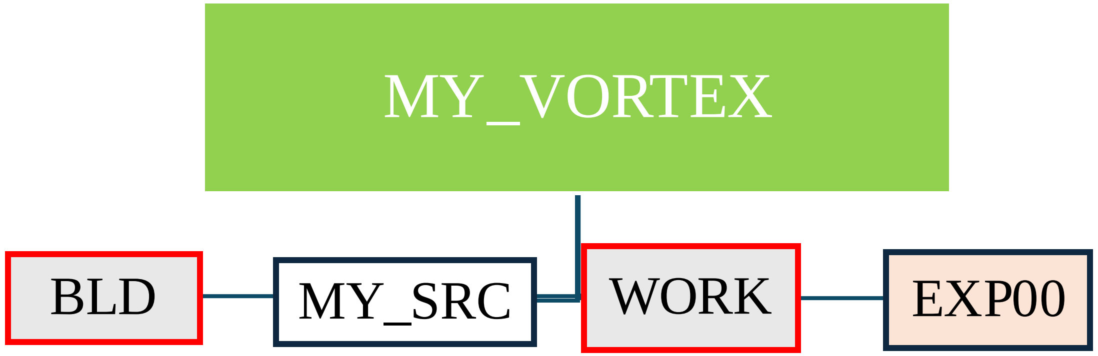


To run your experiment, go in EXP00.
There you find all the needed files.
All the “xml” files are for the management of the model output.
The namelists contain specific model setup.
You need to “submit” you job to the queing system.
I prepared a job submission script

- Cp /home/software/terra/NEMO_env/ancillary/run_nemo_v2.sh

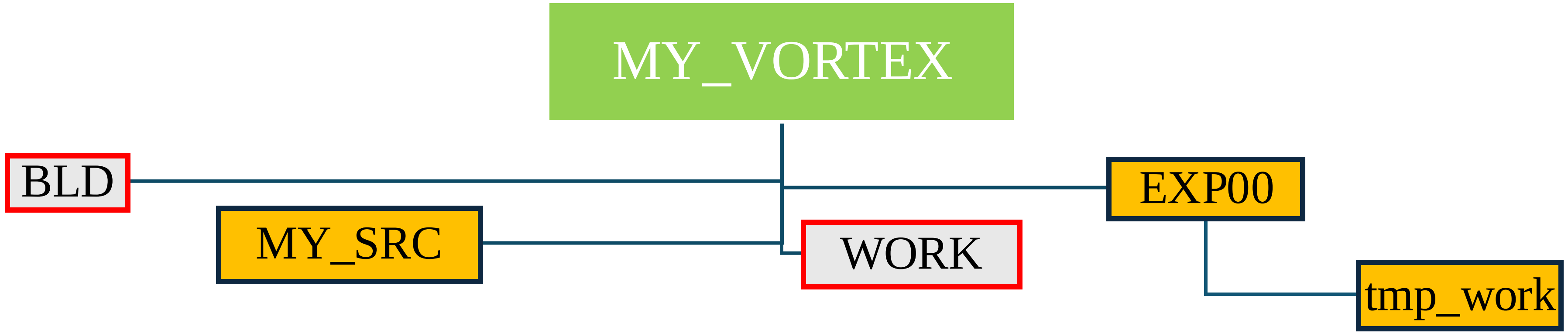
To submit the script
> Sbatch run_nemo_v2.sh

NEMO architecture



- If you modify the *namelists* you can run the code without recompiling.
- The NEMO code will always read the *namelist_cfg*.
- Be sure to make a copy of the original.
- Be sure when submitting the Job that *namelist_cfg* contains the parameters you want to use in your experiment.
- If you want to modify a parameter which is not in *namelist_cfg* you can copy the specific section from *namelist_ref* into your *namelist_cfg* and then modify the parameters

NEMO architecture



Sbatch run_nemo_v2.sh

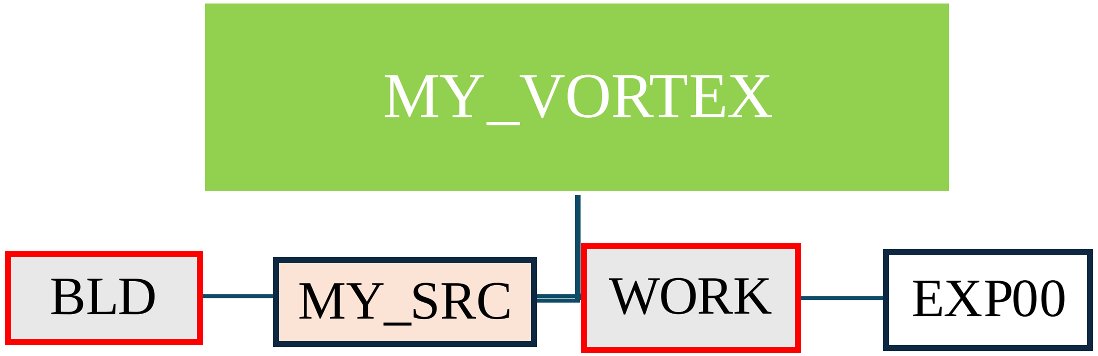
You can control the status of you job

Squeue

The submission script will create a new directory “tmp_work”

In “tmp_work” you will find your model results

NEMO architecture



In MY_SRC there are specific Fortran file defining your experiments

- usrdef_hgr.F90 (horizontal geometry)
- usrdef_istate.F90 (initial condition)
- usrdef_nam.F90 (change in the namelist, add parameters)
- usrdef_sbc.F90 (surface boundary conditions)
- usrdef_zgr.F90 (vertical geometry)

If you modify one of them, you need to recompile the code.

NEMO architecture

In /home/software/terra/NEMO_env/ancillary there are files to help you.

- Compile_nemo.sh there are all the examples on how to compile different test-cases of configurations
- Run_nemo_v2.sh is the script you can you to submit your job
- The directory MY_SRC_EQ_WAVES contains all the fortran files needed to run this specific experiment. You should copy all the files in your MY_SRC directory.
- XML directory contains modified versions of the standard XML. Names should be self-explanatory
- Namelist directory contains some “cfg” namelist used in specific experiments. Name should be self-explanatory.

Example: running the equatorial wave experiment

I assume you already copied the architecture files.

1. Go in nemo directory
2. ./makenemo -m 'OPH' -r GYRE_PISCES -n 'EQ_WAVES' del_key 'key_top key_linssh’

you now have a new experiment called 'EQ_WAVES’ in the cfgs directory

3. Go in the corresponding MY_SRC directory (i.e nemo/cfgs/EQ_WAVES/MY_SRC)
4. Copy the files provided into the home/software/terra/NEMO_env/ancillary/ MY_SRC_EQ_WAVES
5. Go back to the nemo directory and recompile
6. ./makenemo -m 'OPH' -r 'EQ_WAVES’
7. Go to cfgs/EQ_WAVES/EXP00
8. Copy the run_nemo_v2.sh here
9. Copy the eq_waves_namelist_cfg here and re-name as namelist_cfg
10. Copy the eq_waves_file_def_nemo.xml here and re-name as file_def_nemo.xml
11. Copy the submission script and execute sbatch Run_nemo_v2.sh

ERROS

Several kinds of errors may occur.

First check to be done in EXP00/tmp_work check the file ocean.output and look for "E R R O R"
These errors are handled by NEMO

Second check in EXP00 check the files nemo_script.out nemo_script.err